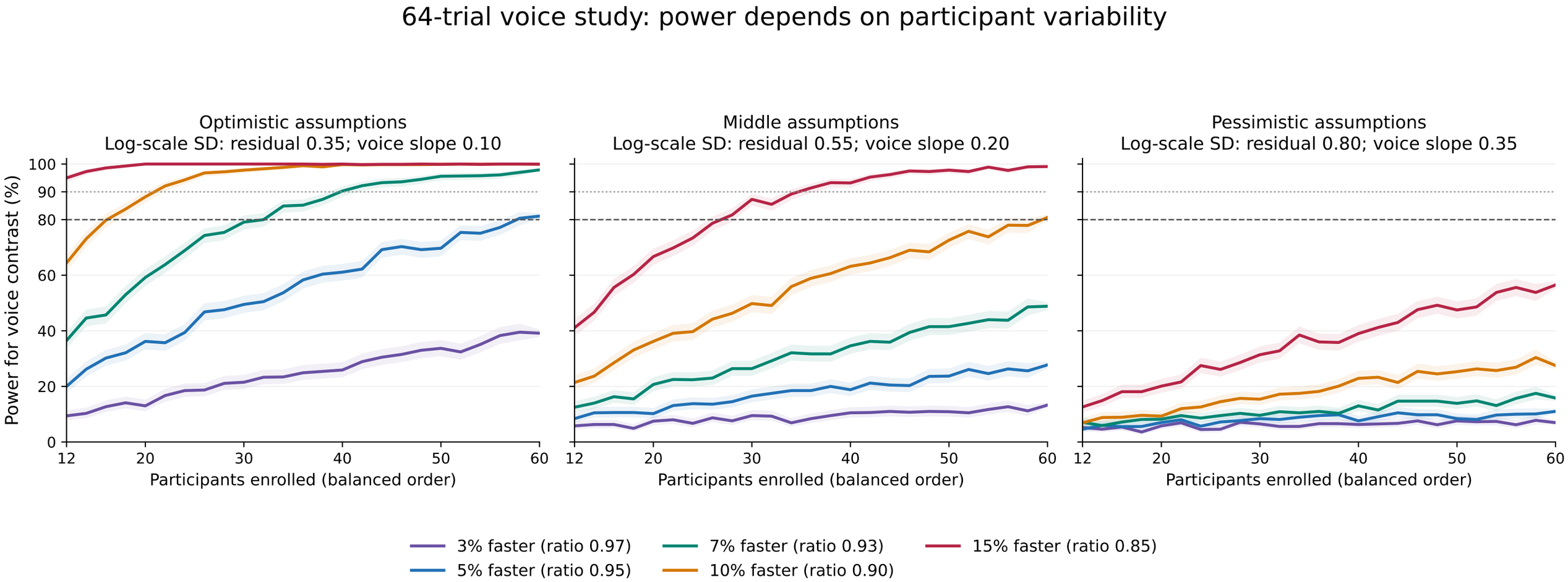
64-trial voice study: power depends on participant variability
Optimistic assumptions
Log-scale SD: residual 0.35; voice slope 0.10
Middle assumptions
Log-scale SD: residual 0.55; voice slope 0.20
Pessimistic assumptions
Log-scale SD: residual 0.80; voice slope 0.35
Power for voice contrast (%)
0
20
40
60
80
90
100
12
20
30
40
50
60
Participants enrolled (balanced order)
3% faster (ratio 0.97)
5% faster (ratio 0.95)
7% faster (ratio 0.93)
10% faster (ratio 0.90)
15% faster (ratio 0.85)
Synthetic planning • 25 s deadline • successful trials only • dropout included • 95% Monte Carlo bands • 1,000 simulations/point; 5,000 at N=60